AI-Based Battery State of Health Prediction for Safer & More Sustainable Energy Systems

Machine Learning Prototype · NASA Battery Dataset · HistGradientBoosting

THE PROBLEM

Batteries power electric vehicles, smartphones, laptops, and renewable energy storage. Yet battery degradation is notoriously hard to monitor accurately. Poor health estimation causes reduced performance, safety risks, premature replacements, and growing electronic waste.

PROPOSED SOLUTION

A machine learning prototype that estimates battery State of Health (SOH) from discharge-cycle behavior — enabling smarter, safer battery management without requiring expensive hardware upgrades.

DATASET & METHOD

→ **Dataset:** NASA Battery Aging Dataset
→ **Target:** SOH = Capacity / 2.0 Ah (nominal)
→ **Filter:** Realistic SOH range [0.5 – 1.2]
→ **Validation:** Held-out battery validation
→ **Model:** HistGradientBoostingRegressor

WHY IT MATTERS

- Battery safety & risk reduction
- Electric vehicle reliability
- Smarter clean energy storage
- Extended battery lifespan
- Reduced electronic waste globally

FUTURE ROADMAP

1. Advanced discharge-curve feature engineering
2. Expand to multiple real-world battery datasets
3. Improve interactive SOH dashboard & deploy online demo
4. Study Battery Management System (BMS) integration
5. Explore pilots, partnerships & commercialization

MODEL PERFORMANCE

MEAN ABSOLUTE ERROR	0.0285
ROOT MEAN SQUARE ERROR	0.0468
R^2 SCORE	0.8303
AVG SOH ERROR	~2.85%
TRAINING SAMPLES	2,307

PROJECT STATUS

Active prototype — Not a commercial product. Built on real NASA data with held-out battery validation.

TECH STACK

Python · Scikit-learn · Pandas · NumPy · Jupyter

Vision: *To develop this prototype into a* ***battery intelligence system*** *that helps companies and communities use batteries more safely, efficiently, and sustainably — accelerating the transition to clean energy.*

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Battery State of Health Prediction · AI · Sustainability · Clean Energy · Battery Intelligence

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